

From Ranks to Smiles

LQD as a monotone transport and pricing engine

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Abstract

For one expiry, a volatility smile is the price-screen image of a probability law. This note builds that law from a single logistic draw and an increasing *rubber ruler*. The ruler says how fast successive probability ranks travel through log-return space. Its speed is positive by construction, a scalar translation makes the normalized asset a martingale, and one endpoint inequality is the exact price of having a finite forward. From there, option pricing becomes the difference of two tail ledgers. The same upper-share ledger also supplies the analytic calibration Jacobian and the natural cross-expiry convex-order test.

This transport view is mathematically identical to the production LQD family, but it does not use the customary log-quantile-density formula as its narrative spine. We derive the density, the no-butterfly result, the moment strip, Lee wing slopes, exact ATM level/skew/curvature identities, calibration controls, and calendar ordering from the transport itself. The worked cases include an exact constant-speed unit check, a production SPX-like fit with 1.341 volatility-basis-point maximum error, an event distribution with two landing zones, and an analytic-versus-finite-difference Jacobian audit. The aim is a lecture that a quant can prove and a trader can use: few symbols, explicit trade tickets, and a sharp distinction between structural guarantees and numerical controls.

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1 One random number, one expiry

Forget the volatility screen for a moment. Imagine a lottery drum containing all ranks from zero to one. We draw one rank p , turn it into a real-valued score, and then decide where that score lands on the log-return axis. If the landing map is increasing, ranks never cross. That innocent observation is the whole architectural idea behind LQD.

Fix an expiry T . Let F_T be its forward price and D_T its discount factor. Expectations are under the T -forward pricing measure, with deterministic proportional carry. Normalize:

$$Y = \frac{S_T}{F_T}, \quad X = \log Y, \quad k = \log \frac{K}{F_T}, \quad \mathbb{E}[Y] = 1. \quad (1)$$

The undiscounted call in forward units is

$$c(k) = \frac{C^{\$}(F_T e^k, T)}{D_T F_T} = \mathbb{E}[(Y - e^k)^+]. \quad (2)$$

Actual dollars return by multiplying by $D_T F_T$. Until then, the forward is one and the strike is e^k .

1.1 The seed

Let Z have the standard logistic distribution

$$\Lambda(z) = \frac{1}{1 + e^{-z}}, \quad \rho(z) = \Lambda(z)(1 - \Lambda(z)). \quad (3)$$

Then $P = \Lambda(Z)$ is uniform on $(0, 1)$ and $Z = \log(P/(1 - P))$. The score z is just log odds: zero is the median, $\log 9$ is the 90th percentile, and $-\log 9$ is the 10th.

Why start here? The logistic coordinate sends both troublesome percentile endpoints to ordinary infinities. More importantly, its density has clean tails:

$$\rho(z) \sim e^z \quad (z \rightarrow -\infty), \quad \rho(z) \sim e^{-z} \quad (z \rightarrow +\infty). \quad (4)$$

Those two exponentials will later turn the model's endpoint speeds directly into moment limits.

Heuristic

Think of Z as a standardized rank, not as a return. The model does not ask whether a move of one score unit is large. It chooses how much log-return distance one score unit should cover at each percentile.

1.2 The speedometer

Choose a smooth log-speed profile h on $[0, 1]$ and set

$$v(z) = \exp(h(\Lambda(z))). \quad (5)$$

The exponential is doing serious work: $v(z) > 0$ for every finite parameter vector. Integrating the speed gives an anchored ruler

$$b(z) = \int_0^z v(t) dt. \quad (6)$$

Finally translate the ruler by a scalar m . The model in this lecture is

$$X = x(Z), \quad x(z) = m + \int_0^z \exp(h(\Lambda(t))) dt, \quad h \in \mathcal{P}_N[0, 1]. \quad (7)$$

This is the note's one central boxed equation. Everything else is a consequence, a numerical method, or a control.

The polynomial is written in shifted Legendre modes,

$$h(p) = \sum_{n=0}^N d_n P_n(1 - 2p). \quad (8)$$

Legendre polynomials are useful rather than sacred: they span the desired low-degree functions, are well conditioned on $[0, 1]$, and let low modes carry broad shape while high modes repair local detail. The construction remains valid with any finite basis of bounded continuous functions having finite endpoint limits. The sharper endpoint expansions used below require the polynomial, or comparable smoothness, assumption. The production dictionary is given in appendix B.

Invariant

The model's primitive object is the positive speed v . A wider band between two neighbouring ranks means less density there; a narrower band means more density. We do not draw volatility and hope it corresponds to probability. We decide how probability ranks travel through return space.

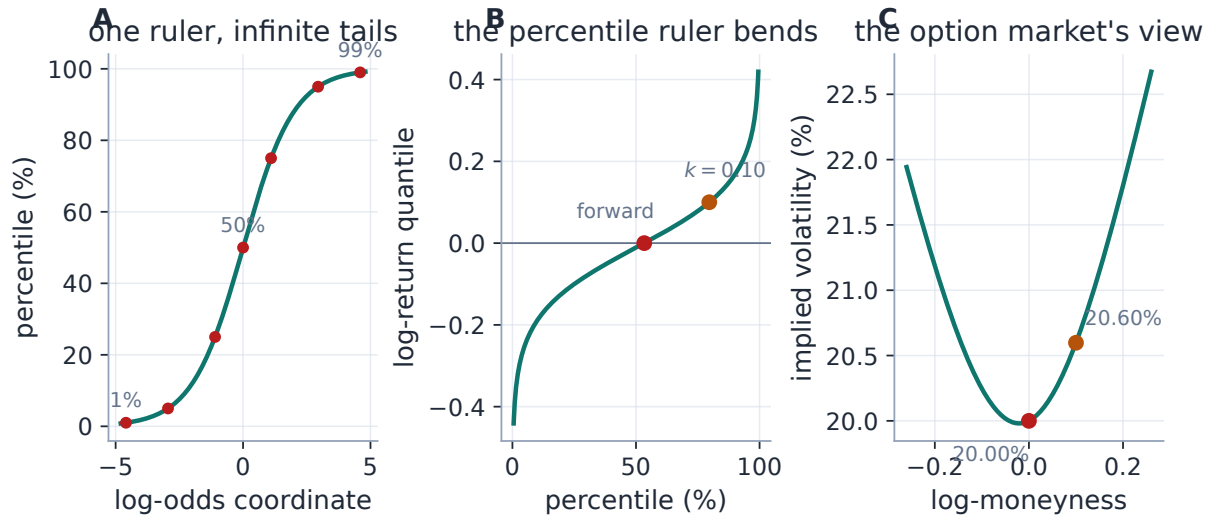


Figure 1: **The percentile ruler in one exact toy.** Left: the logistic score turns percentiles into an unbounded coordinate. Middle: a constant-speed ruler maps that score to log return; the marked forward and 10% log-strike locate the two roots used in pricing. Right: the same law is seen by the desk as an implied-volatility smile. The ATM volatility is exactly 20.0000%, while the $k = 0.10$ point is 20.5977%. One random rank has become a complete option slice.

2 The centering trade and the right-tail credit limit

The raw ruler b has an arbitrary origin. Option pricing does not: in forward units the asset must have mean one. Since $Y = e^{m+b(Z)}$, the unique centering shift is

$$m = -\log \int_{-\infty}^{\infty} e^{b(z)} \rho(z) dz. \quad (9)$$

Whenever this integral is finite,

$$\mathbb{E}[e^X] = e^m \int_{-\infty}^{\infty} e^{b(z)} \rho(z) dz = 1. \quad (10)$$

The shift changes location, not shape: $x'(z) = v(z)$ before and after normalization.

2.1 One hard inequality

The two endpoint speeds are

$$\lambda_- = e^{h(0)}, \quad \lambda_+ = e^{h(1)}. \quad (11)$$

Because h is a polynomial and $\Lambda(z) = O(e^z)$ on the left while $1 - \Lambda(z) = O(e^{-z})$ on the right,

$$b(z) = \lambda_- z + \delta_- + O(e^z), \quad z \rightarrow -\infty, \quad (12)$$

$$b(z) = \lambda_+ z + \delta_+ + O(e^{-z}), \quad z \rightarrow +\infty, \quad (13)$$

for finite constants $\delta_{\pm}$. Combining these expressions with (4), the integrand in (9) behaves as

$$e^{b(z)} \rho(z) \sim \begin{cases} C_- e^{(1+\lambda_-)z}, & z \rightarrow -\infty, \\ C_+ e^{-(1-\lambda_+)z}, & z \rightarrow +\infty. \end{cases} \quad (14)$$

The left tail always integrates because $\lambda_- > 0$. The right tail integrates exactly when $\lambda_+ < 1$.

Proposition 1 (Finite-forward criterion). *The construction in (7) admits a finite martingale shift if and only if*

$$\lambda_+ < 1. \quad (15)$$

At $\lambda_+ = 1$ the normalizer diverges; this is a hard boundary, not a calibration preference.

Proof. Equation (14) makes the left integral finite for every positive λ_- . The right integral is comparable to $\int^{\infty} e^{-(1-\lambda_+)z} dz$, which is finite precisely for $\lambda_+ < 1$. Equation (9) then gives the unique finite translation. $\square$

Caution

Approaching the wall is legal but expensive. As $\lambda_+ \uparrow 1$, far right states carry more and more of the mean. The normalizer, tail quadrature, price evaluation, and parameter sensitivities become numerically ill conditioned before the mathematical boundary is reached. Production therefore adds a soft turn ahead of the hard wall and rejects $\lambda_+ \geq 1 - 10^{-6}$ as a numerical safety buffer. Neither control changes the mathematical boundary at one.

2.2 The density falls out of the ruler

Since $x'(z) = v(z) > 0$ and $x(z) \rightarrow \pm\infty$, the map x is an increasing bijection. Therefore

$$F_X(x(z)) = \Lambda(z), \quad f_X(x(z)) = \frac{\rho(z)}{v(z)} > 0. \quad (16)$$

For the normalized asset,

$$f_Y(e^{x(z)}) = \frac{\rho(z)}{e^{x(z)}v(z)} > 0. \quad (17)$$

Let $\tilde{c}(y) = \mathbb{E}[(Y - y)^+]$ for a strike $y > 0$. Differentiation in strike gives

$$\tilde{c}'(y) = -\mathbb{P}(Y > y), \quad \tilde{c}''(y) = f_Y(y) \geq 0. \quad (18)$$

So the slice is decreasing and convex in *strike*. Convexity is not asserted in log-strike k , where a change of variable contributes extra terms.

Proposition 2 (One-expiry static arbitrage). *If h is finite and $\lambda_+ < 1$, the normalized call curve has the correct boundaries, obeys put–call parity, and is strictly convex at every positive strike. In particular, every continuous butterfly spread has non-negative value.*

Proof. The construction gives $Y > 0$ and $\mathbb{E}[Y] = 1$. Thus $\tilde{c}(0) = 1$, $\tilde{c}(y) \rightarrow 0$ as $y \rightarrow \infty$, and $(1 - y)^+ \leq \tilde{c}(y) \leq 1$. Equation (18) supplies monotonicity and convexity. Applying the same law to puts yields $\tilde{c}(y) - \tilde{p}(y) = 1 - y$. $\square$

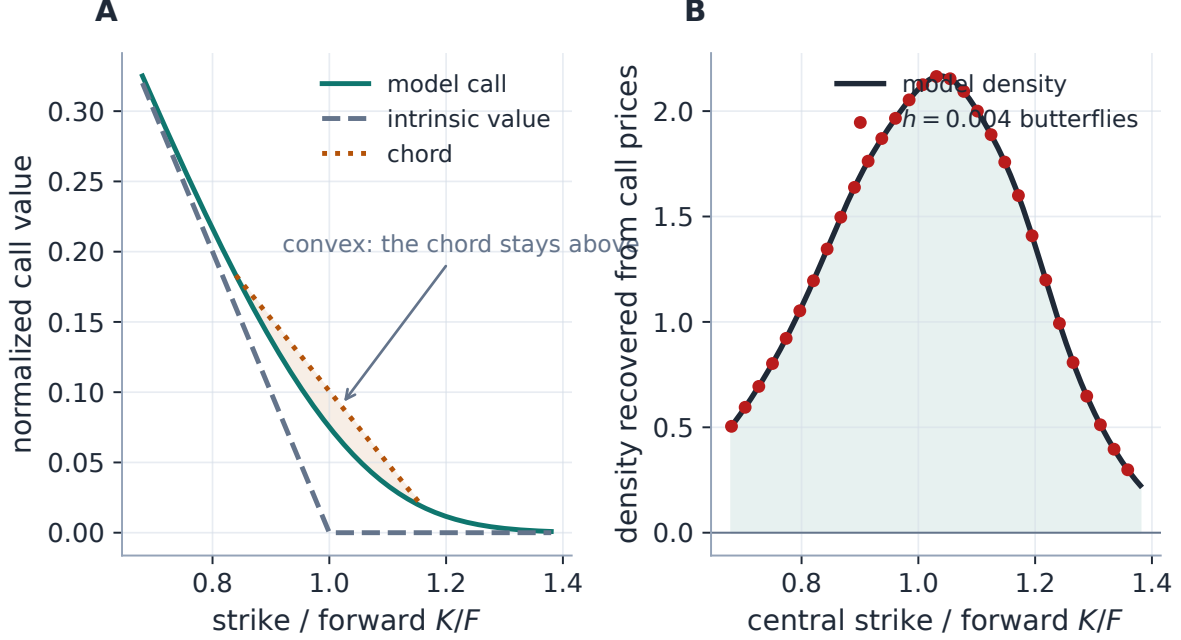


Figure 2: **Butterfly freedom is visible in price space.** The left panel places the call above intrinsic value and below every secant chord required by convexity. The right panel recovers the continuous asset density and compares it with discrete butterfly estimates. In this audit the minimum recovered density over the sampled window is 0.224428, and the maximum relative discretization discrepancy on that grid is 0.0182%.

3 A call is two tail ledgers

The probability law is now complete, but a trader wants a price. Define the upper asset-share ledger

$$G(z) = \int_z^\infty e^{x(t)} \rho(t) dt. \quad (19)$$

It records how much of the mean-one asset is carried by states whose logistic score exceeds z . It starts at one, ends at zero, and satisfies

$$G'(z) = -e^{x(z)} \rho(z). \quad (20)$$

For a log-strike k , let z_k be the unique root

$$x(z_k) = k, \quad p_k = \Lambda(z_k). \quad (21)$$

Splitting the payoff gives

$$\begin{aligned} c(k) &= \int_{z_k}^\infty (e^{x(z)} - e^k) \rho(z) dz \\ &= G(z_k) - e^k (1 - p_k). \end{aligned} \quad (22)$$

This formula has a desk interpretation:

- $G(z_k)$ is the asset delivered in exercised states;
- $e^k(1 - p_k)$ is the strike bill paid in those states.

The call is the asset tail minus the cash tail. No volatility interpolation is involved.

3.1 A complete trade ticket

A six-month call at $k = 0.10$

Take the constant-speed model calibrated to exactly 20.0000% ATM volatility at active variance time $\tau = 0.50$. Its speed is $s = 0.08125025$ and its martingale shift is $m = -0.01088288$. At $k = 0.10$, the exercise percentile is $p_k = 0.796524$ and the normalized strike is $e^k = 1.105171$. The asset ledger is $G(z_k) = 0.24722094$; the cash ledger is $e^k(1 - p_k) = 0.22487593$. Their difference is

$$c(0.10) = 0.24722094 - 0.22487593 = 0.02234501 = 2.2345\% \text{ of forward.}$$

Black inversion reports 20.5977%. The price is primary; the volatility is its market label.

The put ledger is obtained from the lower tail, but it need not be computed separately. Writing its price as $c_{\text{put}}(k)$, mean-one normalization gives

$$c(k) - c_{\text{put}}(k) = 1 - e^k. \quad (23)$$

It also supplies a useful implementation test: price calls from the upper tail, recover puts by parity, and verify that a direct lower-tail calculation agrees.

3.2 Digital and density from the same root

The root motion cancels when differentiating the payoff integral, because the payoff is zero at the exercise boundary. Consequently

$$c'(k) = -e^k(1 - p_k), \quad (24)$$

and one more derivative gives

$$c''(k) = -e^k(1 - p_k) + e^k \frac{\rho(z_k)}{v(z_k)}. \quad (25)$$

The combination

$$e^{-k}(c''(k) - c'(k)) = \frac{\rho(z_k)}{v(z_k)} = f_X(k) \quad (26)$$

recovers the log-return density. In strike coordinates this is the Breeden–Litzenberger identity [1]; in the transport coordinate it is one division.

Heuristic

The function G will recur twice. In section 9, its parameter derivatives give the calibration Jacobian without differentiating z_k . In section 11, G becomes the integrated-quantile curve that orders expiries. Pricing, differentiation, and calendars are three uses of the same ledger.

4 The constant-speed audit

Before trusting a flexible model, make it solve a case that fits on one page. Set

$$h(p) = \log s, \quad 0 < s < 1. \quad (27)$$

Then $v(z) = s$, $b(z) = sz$, and the logistic moment-generating function gives

$$\mathbb{E}[e^{sZ}] = \frac{\pi s}{\sin(\pi s)}. \quad (28)$$

The martingale shift and transport are therefore

$$m = -\log \frac{\pi s}{\sin(\pi s)}, \quad x(z) = m + sz. \quad (29)$$

The condition $s < 1$ is simultaneously the domain of the moment in (28) and the hard right-tail condition (15).

An exact 20% ATM unit check

Solving the one-dimensional equation

$$2\Phi\left(\frac{0.20\sqrt{0.50}}{2}\right) - 1 = c(0; s)$$

gives $s = 0.08125025$. Equation (29) then gives $m = -0.01088288$, and the forward lies at percentile 53.3436%, not at the median. The latter is not a bug: martingale centering occurs in *price*, so the log return usually has a negative location shift.

Quantity	Production value
Solved scale	0.08125025
Martingale shift	-0.01088288
ATM percentile	53.3436%
$C(0.10)$	0.02234501
$\sigma_{BS}(0.10)$	20.5977%

The constant-speed law is not a flat Black smile. It is a log-logistic distribution rescaled to mean one, so its wing moments are finite only over a strip. Its purpose is more valuable than realism: every integration, root-finding, normalization, price, and implied-vol inversion path can be checked against a transparent one-parameter model.

5 The ATM microscope

At the money, three distributional facts replace a neighbourhood of finite differences. Let z_* solve $x(z_*) = 0$ and define

$$p_* = \Lambda(z_*), \quad f_* = \frac{\rho(z_*)}{v(z_*)} = f_X(0). \quad (30)$$

Equations (22)–(25) give

$$c_0 = G(z_*) - (1 - p_*), \quad c'_0 = -(1 - p_*), \quad c''_0 = f_* - (1 - p_*). \quad (31)$$

The ATM price sets level, the exercise percentile sets the digital, and the local compression of the ruler sets density.

5.1 Pass those facts through Black

Let $B(k, w)$ denote the normalized Black call with total variance w , and let $w(k)$ solve

$$B(k, w(k)) = c(k). \quad (32)$$

At $k = 0$, put

$$a = \Phi^{-1}\left(\frac{1 + c_0}{2}\right), \quad w_0 = 4a^2, \quad n = \varphi(a), \quad \Delta = p_* - \Phi(a). \quad (33)$$

Implicit differentiation produces unusually compact identities:

$$w'_0 = \frac{2\sqrt{w_0}}{n} \Delta, \quad (34)$$

$$w''_0 = \frac{2\sqrt{w_0}}{n} f_* - 2 + \left(\frac{1}{2w_0} + \frac{1}{8}\right) (w'_0)^2. \quad (35)$$

The derivation is recorded in appendix A; the interpretation belongs here. Total-variance skew is exactly a *digital mismatch*: p_* is the model percentile of the forward, while $\Phi(a)$ is the corresponding percentile under the flat-Black law with the same ATM option price. Curvature adds the local-density mismatch and a necessary nonlinear skew correction.

If τ is the active variance time and $\sigma(k) = \sqrt{w(k)/\tau}$, then

$$\sigma_0 = \sqrt{\frac{w_0}{\tau}}, \quad (36)$$

$$\sigma'_0 = \frac{\Delta}{n\sqrt{\tau}}, \quad (37)$$

$$\sigma''_0 = \frac{1}{\sqrt{\tau}} \left[\frac{f_*}{n} - \frac{1}{\sqrt{w_0}} + \frac{\sqrt{w_0}}{4} \left(\frac{\Delta}{n}\right)^2 \right]. \quad (38)$$

These are exact identities for the continuous slice. They are semi-analytic, not elementary coefficient formulas: z_* and c_0 still come from monotone inversion and quadrature.

Proposition 3 (ATM information content). *Once the martingale law is built, ATM level, log-strike skew, and log-strike curvature depend only on (c_0, p_*, f_*) and the variance clock τ . No neighbouring implied-volatility quotes are required.*

Proof. At ATM, $B(0, w) = 2\Phi(\sqrt{w}/2) - 1$, which gives (33). The Black partial $B_w(0, w) = \varphi(\sqrt{w}/2)/(2\sqrt{w})$ together with (31) yields (34). Differentiating $B(k, w(k)) = c(k)$ twice and simplifying the ATM Black partials gives (35). The chain rule for $\sigma = \sqrt{w/\tau}$ gives (36)–(38). $\square$

The digital explains skew before any regression does

Suppose the model forward percentile p_* lies below the matched-Black percentile $\Phi(a)$. Then $\Delta < 0$, so (37) says the implied-volatility skew is negative. The statement does not rely on a three-point slope, a fitted parabola, or a small bump size. It is an exact comparison of two digitals.

6 The part of the smile no quote sees

Liquid quotes constrain the body. Extrapolation is governed by what the ruler does as ranks approach zero and one. From (12)–(13),

$$\mathbb{P}(X > x) \sim K_+ e^{-x/\lambda_+}, \quad \mathbb{P}(X < x) \sim K_- e^{x/\lambda_-}, \quad (39)$$

for positive constants $K_{\pm}$. Equivalently, $Y = e^X$ has power tails:

$$\mathbb{P}(Y > y) \sim K_+ y^{-1/\lambda_+}, \quad \mathbb{P}(Y < y) \sim K_- y^{1/\lambda_-}. \quad (40)$$

6.1 Moment budget

The same endpoint comparison gives the complete moment strip

$$\mathbb{E}[e^{rX}] < \infty \iff -\frac{1}{\lambda_-} < r < \frac{1}{\lambda_+}. \quad (41)$$

For the asset Y , the critical Lee moments are

$$\pi_+ = \frac{1}{\lambda_+} - 1, \quad \pi_- = \frac{1}{\lambda_-}. \quad (42)$$

The first says how many moments beyond the forward survive on the right; the second says how many inverse moments survive on the left.

Lee's moment formula [2] uses the map

$$\Psi(q) = 2 - 4 \left(\sqrt{q^2 + q} - q \right). \quad (43)$$

For total implied variance, the wing limsups are

$$\limsup_{k \rightarrow \infty} \frac{w(k)}{k} = \Psi(\pi_+), \quad \limsup_{k \rightarrow -\infty} \frac{w(k)}{|k|} = \Psi(\pi_-). \quad (44)$$

Substituting (42) yields direct speed-to-slope formulas:

$$\beta_+ = \frac{2\lambda_+}{(1 + \sqrt{1 - \lambda_+})^2}, \quad 0 < \lambda_+ < 1, \quad (45)$$

$$\beta_- = \frac{2\lambda_-}{(1 + \sqrt{1 + \lambda_-})^2}, \quad \lambda_- > 0. \quad (46)$$

For small endpoint speeds, $\beta_{\pm} \sim \lambda_{\pm}/2$. As the right speed approaches the hard wall, $\beta_+ \rightarrow 2$.

Caution

Lee's theorem itself gives the limsup statement (44). The polynomial endpoint expansion here produces regular power tails, so regular-variation tail-wing refinements [3] upgrade the wing rates to ordinary limits. When only Lee's moment result is being invoked, the careful word is *limsup*.

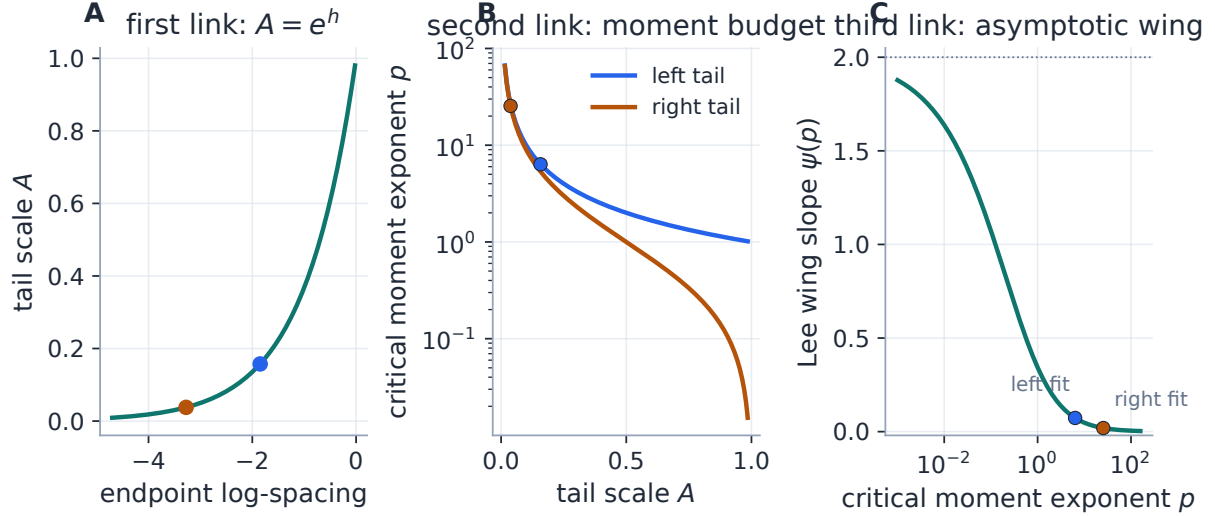


Figure 3: **The endpoint is an extrapolation forecast.** Left: the speed profile settles to two endpoint values. Centre: those values are inverse moment budgets through (42). Right: the moment budgets map to Lee total-variance slopes. The marked production fit has $(\lambda_-, \lambda_+) = (0.157473, 0.037786)$ and $(\beta_-, \beta_+) = (0.073087, 0.019259)$.

The right endpoint has a balance-sheet meaning

For the SPX-like fit in section 7, the right speed is $\lambda_+ = 0.037786$. Hence the model retains moments $\mathbb{E}[Y^{1+q}]$ for $q < 1/0.037786 - 1$ and predicts a right-wing Lee slope of 0.019259. Moving a high-order coefficient can leave quoted strikes nearly unchanged yet alter $h(1)$, the critical moment boundary, and the far wing. A small in-window residual is therefore not an extrapolation diagnostic.

7 Put the quote screen back in

Only now do we return to market quotes. The model provides arbitrage-safe prices; Black inversion merely expresses those prices in familiar units. For total variance w , the normalized Black call is

$$B(k, w) = \Phi(d_+) - e^{-k} \Phi(d_-), \quad d_{\pm} = -\frac{k}{\sqrt{w}} \pm \frac{\sqrt{w}}{2}. \quad (47)$$

At quote i , target price $B(k_i, w_i^{\text{mkt}})$ is compared with model price $c(k_i; \theta)$. A useful core residual is

$$r_i(\theta) = \omega_i^{1/2} \frac{c(k_i; \theta) - B(k_i, w_i^{\text{mkt}})}{\text{Vega}_i + \eta}. \quad (48)$$

Here $\eta > 0$ prevents tiny far-wing vegas from acquiring absurd weight. With normalized Black vega $\text{Vega}_i = \varphi(d_{+,i})\sqrt{\tau}$, r_i is approximately an implied-volatility error.

7.1 What the objective is allowed to prefer

The residual stack adds controls, each with a distinct economic meaning:

- bid–ask bands can replace point targets with zero hinge cost inside the spread, while retaining a gentle midpoint anchor;
- a ridge on modes $n \geq 4$ discourages oscillatory speed profiles that sparse quotes cannot identify;
- a one-sided soft barrier turns the optimizer away before $\lambda_+ = 1$;
- optional priors stabilize weakly identified tails or preserve continuity with the previous fit;
- optional calendar rows penalize crossings of upper-share ledgers;
- optional log-contract and ATM/RR/BF operator rows express desk views.

The hard martingale wall remains hard. Every other row is a weighted modelling choice and should be reported as such.

Performance

Production prices in the native coordinate. It tabulates $x(z)$ and $G(z)$ once, solves all strike roots monotonically, and evaluates each call from (22). This is faster and numerically calmer than integrating each payoff from scratch.

7.2 A deterministic SPX-like approximation case

The first stress case uses a smooth skewed target sampled at 24 strikes and fits an order-9 LQD slice with the production calibrator. It is a capacity and implementation test, not a claim about a live market or a trading signal.

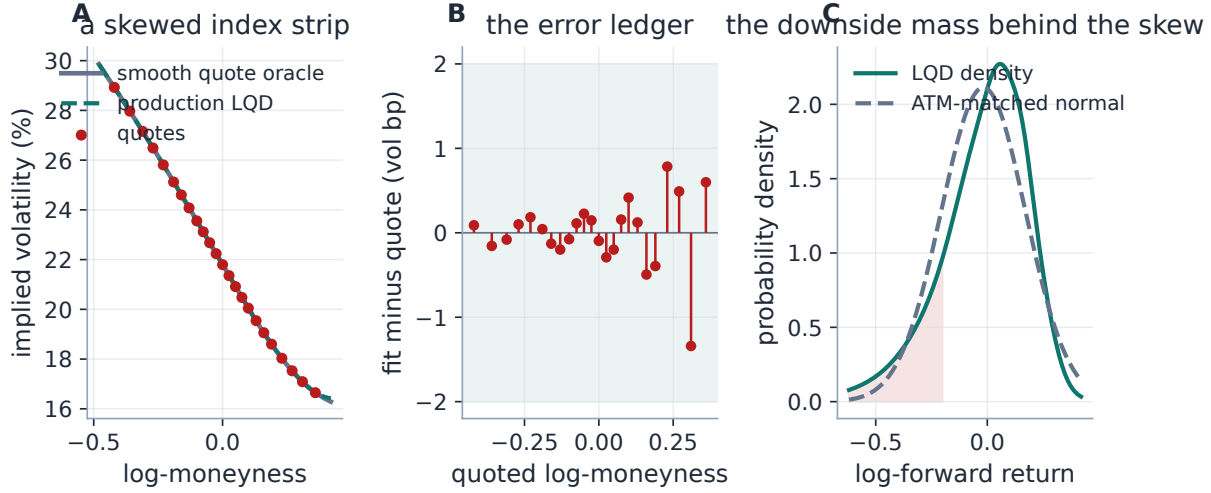


Figure 4: **A quote screen, its residual ledger, and the law underneath.** Left: the order-9 production fit overlays the deterministic SPX-like target. Centre: all residuals remain inside a narrow volatility-basis-point band; maximum and RMS errors are 1.341 and 0.408 bp. Right: the fitted return density is compared with an ATM-matched normal, making the downside allocation of probability visible.

Read the fit in three coordinate systems

The optimizer terminates in 8 function evaluations. The quote screen reports ATM volatility 21.7961%, skew -0.177846, and curvature 0.002673. The distribution reports endpoint speeds (0.157473, 0.037786) and martingale error -1.11×10^{-16} . The terminal log contract reports a log-contract-implied volatility of 22.9685%. None of these views replaces the others: quotes measure fit, endpoint speeds measure extrapolation, and the density shows where probability actually landed.

8 How to move level without wrecking shape

Raw polynomial coefficients are efficient calibration coordinates and poor conversation. A trader asks for more level with skew unchanged, not for a bump to the fourth Legendre coefficient. Let

$$H(\theta) = \begin{pmatrix} w_0(\theta) \\ \sigma'_0(\theta) \\ \sigma''_0(\theta) \end{pmatrix}, \quad D = \frac{\partial H}{\partial \theta}. \quad (49)$$

The internal production chart uses w_0 in its first coordinate. The screen-facing level is σ_0 , and at fixed τ the two charts are locally equivalent through $w_0 = \tau \sigma_0^2$. At a regular point, D has row rank three. The minimum-norm first-order move that changes handles by δH is

$$\delta \theta_{\text{primary}} = D^\top (D D^\top)^{-1} \delta H. \quad (50)$$

Any vector in $\ker D$ is a first-order shape move: it changes the smile while leaving ATM level, skew, and curvature fixed to first order.

Assumption 1 (Regular handle point). The parameter vector lies strictly inside $\lambda_+ < 1$, the ATM option has non-negligible vega, and D has row rank three. Near a rank loss, the pseudoinverse is a warning signal rather than a trading control.

The linear move is local. A finite requested shift can be attempted with Newton retargeting on the exact handles (36)–(38). The current routine is not a globally feasible map: an intermediate iterate can meet the martingale wall, and the solve can fail near a singular chart. This distinction matters: a linear chart is a compass, not a teleport.

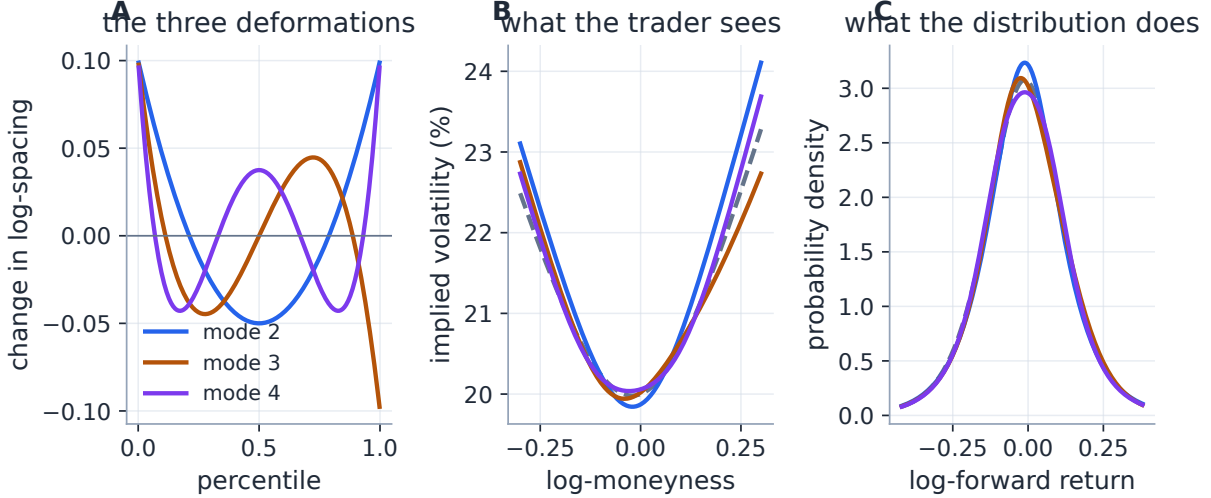


Figure 5: **What low polynomial modes actually move.** Each panel is one view — log-speed, implied volatility, or return density — and overlays the same three controlled perturbations. Broad modes move level or tilt; higher curvature redistributes probability more locally. The panels explain why coefficient orthogonality is useful but does not make coefficients into trader handles.

Primary move versus shape move

With order $N = 9$, there are ten polynomial coefficients but only three ATM handles. Under assumption 1, the primary subspace has dimension three and the local shape nullspace has dimension seven. A calibration can therefore match the same ATM level, skew, and curvature with materially different wing or density shapes. Quotes and regularization decide which member of that seven-dimensional family is acceptable.

9 The root that disappears

Calibration speed depends on derivatives. A naive implementation bumps every coefficient, rebuilds the martingale shift, resolves every strike root, and reprices every option. The transport reveals a cleaner route.

Write the profile in any affine coefficient chart,

$$h_{\theta}(p) = \sum_j \theta_j \phi_j(p). \quad (51)$$

The raw-ruler sensitivity is

$$r_j(z) = \frac{\partial b(z)}{\partial \theta_j} = \int_0^z v(t) \phi_j(\Lambda(t)) dt. \quad (52)$$

Differentiate the mean-one identity. Since $1 = \int e^{x(z)} \rho(z) dz$,

$$\frac{\partial m}{\partial \theta_j} = - \int_{-\infty}^{\infty} e^{x(t)} r_j(t) \rho(t) dt, \quad (53)$$

and hence

$$x_j(z) := \frac{\partial x(z)}{\partial \theta_j} = r_j(z) - \int_{-\infty}^{\infty} e^{x(t)} r_j(t) \rho(t) dt. \quad (54)$$

Define the corresponding upper-share sensitivity

$$G_j(z) = \int_z^{\infty} e^{x(t)} x_j(t) \rho(t) dt. \quad (55)$$

Theorem 1 (Envelope cancellation). *At fixed log-strike k and an interior admissible parameter vector,*

$$\frac{\partial c(k; \theta)}{\partial \theta_j} = G_j(z_k). \quad (56)$$

The derivative of the implicit strike root z_k is not needed.

Proof. Differentiate (22). Terms involving the moving root combine to

$$\left[G'(z_k) + e^k \rho(z_k) \right] \frac{\partial z_k}{\partial \theta_j}.$$

But (20) and $x(z_k) = k$ imply $G'(z_k) = -e^k \rho(z_k)$. The bracket vanishes. The remaining fixed-boundary term is exactly (55). $\square$

This is an envelope theorem in trading clothes. Moving the exercise boundary has no first-order value because the option payoff is zero at that boundary. The expensive implicit derivative disappears for an economic reason, not an algebraic accident.

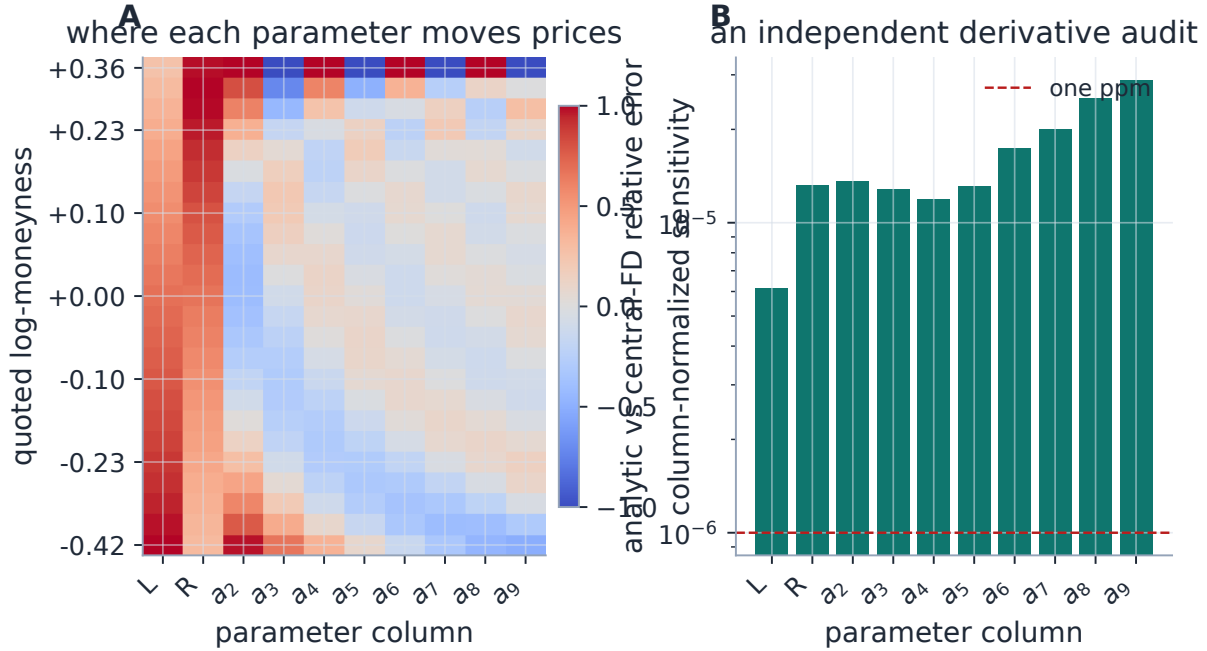


Figure 6: **The analytic price Jacobian under audit.** Left: the production sensitivities for 10 parameters are compared with central finite differences across strikes. Right: normalized column errors remain below 2.88×10^{-5} . The residual discrepancy is the expected finite-grid/interpolation error, not a missing strike-root term.

Caution

The continuous identity is exact under dominated differentiation in an interior neighbourhood. Production tabulates and interpolates x and G separately, so off-grid algebra is not bitwise exact. Sensitivities also grow poorly conditioned near $\lambda_+ = 1$. Finite-difference audits remain a necessary test even when finite differences are no longer the main algorithm.

10 When probability chooses two landing zones

Positive density does not mean unimodal density. If the ruler moves slowly in two separated regions, probability ranks bunch into two return zones. This is useful around discrete events and dangerous when caused by overfitting.

To make that distinction visible, the second deterministic case starts from an asymmetric two-component event law. Its left component carries 56.0% weight and is centred at -0.074804; the right component is centred at 0.085196. We sample 37 option quotes and fit the highest supported production order, $N = 16$.

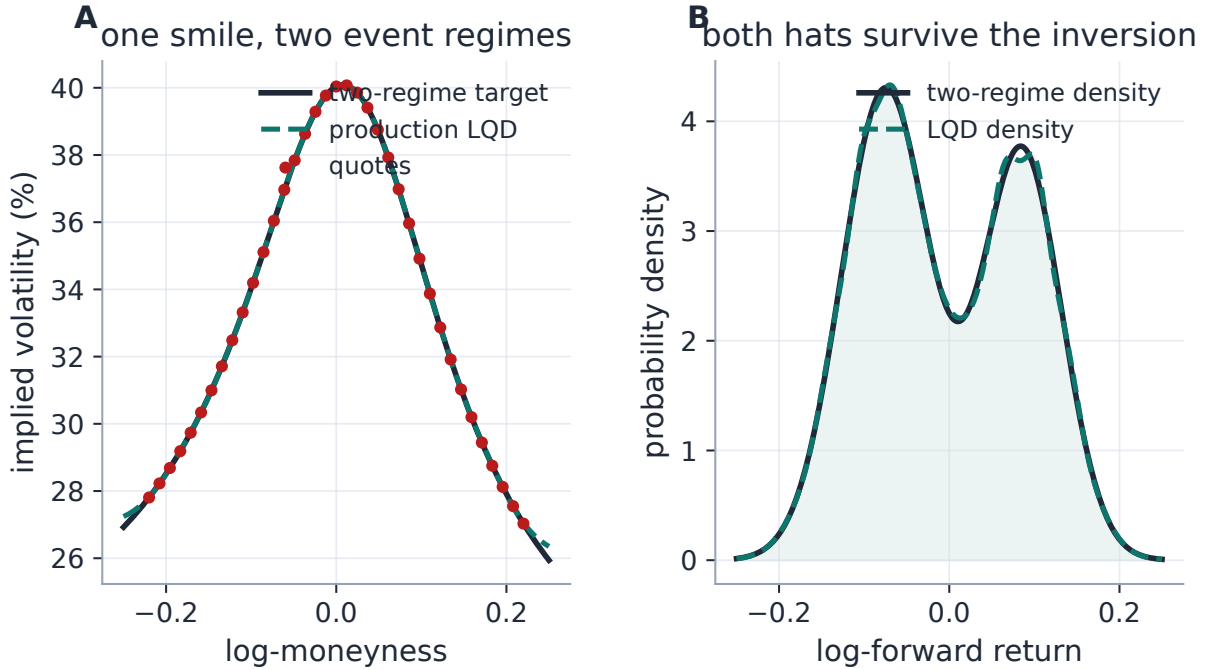


Figure 7: **An adversarial event case.** Left: the order-16 production slice approximates the event-shaped implied-volatility target with maximum and RMS quote errors 2.444 and 0.999 volatility bp. Right: the recovered density visibly selects two landing zones. The fit tests whether a positive transport family can express bimodality; it does not certify that the second mode is economically real.

Positive is guaranteed; plausible is judged

The fit has endpoint speeds (0.015531, 0.014044) and martingale error -2.22×10^{-16} . Its density cannot be negative, regardless of order. Yet a second mode may reflect a genuine scheduled jump, sparse quotes, a loose high-order ridge, or stale prices. Static arbitrage supplies a floor for model quality, not a complete economic opinion.

The example also explains why order is a risk control. Raising N expands approximation capacity and creates more ways to hide oscillation between quotes. High order is justified when the data or prior information can identify the extra structure. Otherwise it is merely freedom with a polished residual.

11 Stacking expiry engines

Each expiry engine is butterfly-free on its own. A surface must also avoid a shorter-dated call being more expensive than a longer-dated call at the same normalized strike. The right object is not the log-return mean or variance; it is convex order for the normalized asset.

Under deterministic proportional carry and a consistent mean-one forward normalization, write for expiry T the percentile map

$$y_T(p) = \exp\left(x_T\left(\log \frac{p}{1-p}\right)\right), \quad 0 < p < 1, \quad (57)$$

and its upper integrated percentile curve

$$\mathcal{G}_T(p) = \int_p^1 y_T(r) dr. \quad (58)$$

The substitution $r = \Lambda(z)$ reveals the promised identity

$$\mathcal{G}_T(p) = G_T\left(\log \frac{p}{1-p}\right). \quad (59)$$

The price ledger is already the calendar ledger.

11.1 Why ordering the ledgers orders every call

At normalized strike y ,

$$\tilde{c}_T(y) = \sup_{0 \leq p \leq 1} \{\mathcal{G}_T(p) - (1-p)y\}. \quad (60)$$

A maximizing percentile is associated with the exercise threshold $y_T(p) = y$. Conversely,

$$\mathcal{G}_T(p) = \inf_{y \geq 0} \{\tilde{c}_T(y) + (1-p)y\}. \quad (61)$$

These conjugate relations prove the ordering result directly; they are the integrated-quantile form of standard convex order [4, 5].

Theorem 2 (Calendar order in the native ledger). *Let Y_1 and Y_2 be positive integrable variables with mean one, and for a general law let $y_i(p)$ denote its left-continuous generalized quantile. Then*

$$Y_1 \leq_{\text{cx}} Y_2 \iff \mathcal{G}_1(p) \leq \mathcal{G}_2(p) \text{ for every } p \in [0, 1]. \quad (62)$$

Equivalently, every normalized call on Y_1 is no more expensive than the same-strike call on Y_2 .

Proof. If the integrated percentiles are ordered, taking suprema in (60) orders every call. If every call is ordered, taking infima in (61) orders every integrated percentile. Equal means supply equality at $p = 0$, and both curves vanish at $p = 1$. $\square$

A calendar crossing can hide between strikes

Suppose two adjacent expiries satisfy $\mathcal{G}_1(p_j) \leq \mathcal{G}_2(p_j)$ at the quoted exercise percentiles but cross between p_j and p_{j+1} . Quote-by-quote checks pass, yet theorem 2 implies that some unquoted strike has a negative calendar spread. A dense logit grid makes such a crossing harder to miss; only a continuum certificate rules it out mathematically.

Production evaluates (59) on a finite logit grid and adds finite-weight hinge residuals for negative gaps. This is effective calendar *control*, not a structural guarantee. It protects listed expiries at sampled ranks; it does not certify every rank, arbitrary interpolation between expiries, or models with inconsistent forward measures.

11.2 The native log contract

Because $P = \Lambda(Z)$ is uniform,

$$\mathbb{E}[X] = \int_0^1 x \left(\log \frac{p}{1-p} \right) dp. \quad (63)$$

For a mean-one positive asset, define the terminal log-contract quantity

$$w_{\log} = -2\mathbb{E}[X]. \quad (64)$$

This route integrates the constructed law directly. It avoids extrapolating a finite strike grid before evaluating the familiar option-strip replication [8]. The annualized display uses the active variance clock, $\sigma_{\log} = \sqrt{w_{\log}/\tau}$. The identity is exact for the terminal log contract. It equals the fair pathwise realized-variance strike under the usual continuous-path replication assumptions; jumps introduce a correction, and a terminal marginal alone cannot price realized variance.

12 The honest contract

A good model note should end its sales pitch before the reader has to. The table below separates what follows from the mathematics from what is merely encouraged by a calibration objective.

Table 1: What the model promises, controls, and does not claim.

Property	Status	Precise statement
Positive continuous density	Structural	$v > 0$ makes x increasing and (16) positive.
Mean-one normalized asset	Structural, conditional	Exact after (9), provided $\lambda_+ < 1$.
One-expiry butterfly freedom	Structural	Calls are convex in normalized strike, not necessarily in log-strike.
Tail moments and Lee rates	Structural	Endpoint speeds determine (41) and (45)–(46).
ATM level, skew, curvature	Exact slice identities	Equations (36)–(38); numerical quadrature and root solves are still required.
Calendar freedom	Soft in production	Exact if (62) holds for all ranks; production samples finitely and uses finite weights.
Unimodality or economic plausibility	Not guaranteed	Positive density can be oscillatory or multimodal.
Atoms and default mass	Not represented directly	A finite smooth speed produces a continuous positive law on $(0, \infty)$.
Parameter identification	Not guaranteed	Sparse quotes may leave tail and high-order directions weakly identified.
Dynamic arbitrage under stochastic carry	Outside the slice theorem	Marginal normalized convex order is not a full dynamic market model.

Invariant

The durable mental model is short. Draw a logistic rank. Move it through a positive-speed ruler. Translate the ruler so the asset mean is one. Price a call as an upper asset share minus an upper cash bill. Read extrapolation from the two endpoint speeds. Everything else is implementation, calibration, or governance.

A Numerical construction and exact ATM algebra

A.1 A stable finite grid

Production works on a finite logistic interval $[-z_{\max}, z_{\max}]$. On a sorted grid z_j , it evaluates

$$p_j = \Lambda(z_j), \quad v_j = e^{h(p_j)},$$

and integrates v outward from the anchor $z = 0$ to obtain b_j . The martingale integrand and upper-share integrand begin with the same array,

$$a_j = e^{b_j} \rho(z_j).$$

The normalizer is $M = \int a(z) dz$, the shift is $m = -\log M$, and the normalized upper share is

$$G(z) = \frac{\int_z^\infty a(t) dt}{M}. \quad (65)$$

Computing numerator and denominator from the unshifted integrand is useful: their potentially large common scale cancels.

The omitted tails are not set to zero. From (12)–(13), their leading integrals are exponential and can be added analytically. For example, at a right boundary z_R ,

$$\int_{z_R}^\infty e^{b(z)} \rho(z) dz \approx \frac{e^{b(z_R)-z_R}}{1 - \lambda_+}, \quad (66)$$

while at a left boundary $z_L < 0$,

$$\int_{-\infty}^{z_L} e^{b(z)} \rho(z) dz \approx \frac{e^{b(z_L)+z_L}}{1 + \lambda_-}. \quad (67)$$

These are the leading production corrections. They become ill conditioned near the hard wall, as they should.

A.2 Hermite interpolation and root solves

The tabulated $x_j = m + b_j$ is strictly increasing. Production uses piecewise cubic Hermite interpolants supplied with the analytic nodal slopes $x' = v$ and $G' = -e^x \rho$. This gives fourth-order local accuracy, but it is not a Fritsch–Carlson monotonicity guarantee [7]; monotonicity between nodes is therefore a numerical audit on the chosen grid. Strike inversion starts from a monotone linear-table seed and applies a small fixed number of clipped Newton updates.

The finite-grid price is then (22). Three routine audits catch most errors:

1. verify $G(-z_{\max})$ plus its left correction is one to tolerance;
2. compare direct lower-tail puts with calls recovered from parity;
3. refine the grid and confirm that prices, densities, and Jacobians converge together.

A.3 ATM Black derivatives

For completeness, let $s = \sqrt{w}$ and $d_{\pm} = -k/s \pm s/2$. At fixed w ,

$$B_k = -e^k \Phi(d_-), \quad B_w = \frac{\varphi(d_+)}{2s}. \quad (68)$$

At $k = 0$, with $a = s/2$ and $n = \varphi(a)$,

$$B_k = \Phi(a) - 1, \quad B_w = \frac{n}{2s}, \quad (69)$$

$$B_{kk} = \frac{n}{s} + \Phi(a) - 1, \quad B_{kw} = \frac{n}{4s}, \quad (70)$$

$$B_{ww} = -n \left(\frac{1}{16s} + \frac{1}{4s^3} \right). \quad (71)$$

Implicit differentiation gives

$$w'_0 = \frac{c'_0 - B_k}{B_w}, \quad (72)$$

$$w''_0 = \frac{c''_0 - B_{kk} - 2B_{kw}w'_0 - B_{ww}(w'_0)^2}{B_w}. \quad (73)$$

Substituting (31) and (69)–(71) simplifies to (34)–(35). The terms linear in w'_0 cancel.

A.4 Two clocks

T labels the expiry and governs discounting and forwards. The quantity τ annualizes total variance. In the base configuration they coincide as year fractions. With an event clock, τ includes the scheduled-event dilation used elsewhere in Vol-Fitter. All Black vegas and annualized handles must use the same active τ that produced the target total variances.

B Production dictionary, controls, and traceability

B.1 How the transport becomes log quantile density

The lecture used h as an ordinary polynomial. To recover the conventional name, define the return quantile at percentile p :

$$Q(p) = x \left(\log \frac{p}{1-p} \right). \quad (74)$$

Since $dz/dp = 1/[p(1-p)]$,

$$Q'(p) = \frac{e^{h(p)}}{p(1-p)}, \quad \log Q'(p) = h(p) - \log p - \log(1-p). \quad (75)$$

This is why the production model is called LQD. In this lecture it is a derived identity, not the starting metaphor.

Write the shifted-Legendre expansion as in (8). The production parameterization separates a linear endpoint chord:

$$h(p) = (1 - p)L + pR + \sum_{n=2}^N a_n P_n(1 - 2p). \quad (76)$$

The charts are exactly equivalent:

$$d_0 = \frac{L + R}{2}, \quad d_1 = \frac{L - R}{2}, \quad d_n = a_n \quad (n \geq 2), \quad (77)$$

and $L = d_0 + d_1$, $R = d_0 - d_1$. Note that L and R are chord coefficients, not by themselves the true endpoint log-speeds. Those are

$$\log \lambda_- = L + \sum_{n=2}^N a_n, \quad (78)$$

$$\log \lambda_+ = R + \sum_{n=2}^N (-1)^n a_n. \quad (79)$$

Table 2: Transport notation and production objects.

Lecture object	Conventional symbol	Production object
$p = \Lambda(z)$	percentile u	<code>slice.u</code>
$z = \text{logit } p$	logit coordinate z	<code>slice.z</code>
$h(p)$	smooth part $g(u)$	basis evaluation
$v(z) = e^{h(\Lambda(z))}$	dQ/dz	<code>slice.dq_dz</code>
$b(z)$	anchored raw quantile	integration workspace
m	martingale shift μ	<code>slice.mu</code>
$x(z)$	normalized quantile on z	<code>slice.q_z</code>
$G(z)$	upper asset share $A(z)$	<code>slice.a_z</code>
λ_-	A_L	<code>a_left</code>
λ_+	A_R	<code>a_right</code>

B.2 Control atlas

Table 3: Principal production controls. Defaults belong to code and configuration; the table records semantics rather than freezing numbers.

Control	Type	Role
order	integer, 4–16	Degree N of h ; raises both capacity and oscillation risk.
z_max , grid size	numerical	Logistic truncation and resolution for integration, inversion, and interpolation.

Control	Type	Role
quote weights, vega floor	objective	Define the price residual metric in (48).
bid-ask mode	objective	Gives zero hinge cost inside a supplied band while retaining the gentle production midpoint anchor.
high-mode ridge	regularizer	Penalizes weakly identified coefficients, normally beginning at degree four.
right-tail turn and weight	soft barrier	Discourages λ_+ before the hard martingale wall at one.
parameter prior	regularizer	Stabilizes tails, warm starts, or continuity when data are sparse.
ATM/RR/BF operator targets	optional views	Add quote-operator prior rows. Exact level/skew/curvature retargeting is the separate local Newton chart of section 8.
calendar grid and weight	soft surface control	Penalize negative adjacent-expiry gaps in (59).
log-contract target	optional view	Target (64) without strike-strip extrapolation.
optimizer tolerances	numerical	Stop rules and evaluation budgets; they do not change feasibility.

B.3 Fresh deterministic benchmark ledger

Table 4: Diagnostics regenerated with the current production implementation.

Diagnostic	SPX-like	Event
Model order	9	16
Quotes	24	37
Max error (vol bp)	1.341	2.444
RMS error (vol bp)	0.408	0.999
A_L	0.157473	0.015531
A_R	0.037786	0.014044

Table 5: Claim-to-implementation traceability.

Mathematical object	Production area	Audit
Legendre log-speed (76)	model parameter/basis code	endpoint values and basis recurrence tests
Integrated transport (7)	slice builder	monotonicity and grid-refinement tests
Martingale shift (9)	normalization path	$ \mathbb{E}[Y] - 1 $ and tail-correction audit
Upper-share price (22)	pricer	parity, intrinsic bounds, quadrature comparison
Density (16)	density API	normalization and discrete-butterfly comparison

Mathematical object	Production area	Audit
ATM handles	handle evaluator	analytic formulas versus centred quote-grid differences
Moment/Lee map	diagnostics	endpoint perturbations and asymptotic identities
Analytic Jacobian (56)	calibration Jacobian	central finite differences; see fig. 6
Calendar ledger (59)	joint residual stack	sampled adjacent-expiry gap reports
Log contract (64)	log-contract view	direct law integral versus option-strip benchmark

C Performance notes and failure modes

C.1 Work that is shared

For one parameter vector, the expensive objects are slice objects, not quote objects:

1. evaluate h and v on the logistic grid;
2. integrate once for b , normalize once for m , and accumulate once for G ;
3. solve all monotone strike roots;
4. evaluate prices and, when requested, all parameter-share ledgers G_j .

Basis values, integration weights, and strike-independent interpolation structures can be cached. Warm starts across neighbouring expiries and market updates often save more time than a more exotic optimizer.

C.2 Numerical qualifications

- **Near-wall conditioning.** Tail corrections contain $1/(1 - \lambda_+)$ and sensitivities can explode before feasibility fails.
- **Interpolation consistency.** Separate Hermite interpolants for x and G are high-order accurate but do not preserve their continuous derivative identity exactly between nodes.
- **Implied-vol inversion.** Deep intrinsic options should be inverted from time value with robust price bounds; tiny vega magnifies price noise.
- **High-order cancellation.** Large alternating coefficients can create moderate h in the quote window and extreme endpoints. Inspect endpoint speeds directly.
- **Grid convergence.** A good quote residual on one grid is not a numerical certificate. Refit or at least reprice on a refined grid.
- **Fallback derivatives.** Core price, band, ridge, barrier, and calendar rows admit analytic or elementary derivatives. Optional blocks not covered by the analytic path may still trigger finite differences.

Performance

The fastest trustworthy Jacobian is not the one with the fewest floating-point operations. It is the one whose columns can be reproduced by finite differences on a refined grid, whose tail sensitivity remains reported near the wall, and whose residual blocks declare when they fall back to numerical bumps.

D Compact reference implementation

The following code is intentionally pedagogical. It displays the transport, martingale normalization, upper-share ledger, and call formula without production caching, tail corrections, or defensive error handling.

Listing 1: A minimal percentile-ruler slice and call pricer.

```
1 import numpy as np
2 from scipy.integrate import cumulative_trapezoid
3 from scipy.interpolate import PchipInterpolator
4 from scipy.special import expit
5
6
7 def build_slice(z, coeff):
8     """Return x(z), G(z), and endpoint speeds."""
9     p = expit(z)
10    xi = 1.0 - 2.0 * p
11    h = np.polynomial.legendre.legval(xi, coeff)
12    speed = np.exp(h)
13
14    # Integrate from the left, then re-anchor at z = 0.
15    raw = cumulative_trapezoid(speed, z, initial=0.0)
16    raw -= np.interp(0.0, z, raw)
17
18    rho = p * (1.0 - p)
19    unshifted_share = np.exp(raw) * rho
20    mass = np.trapezoid(unshifted_share, z)
21    shift = -np.log(mass)
22    x = shift + raw
23
24    share_density = np.exp(x) * rho
25    rev = cumulative_trapezoid(
26        share_density[::-1], z[::-1], initial=0.0
27    )
28    upper_share = -rev[::-1]
29
30    left_speed = np.exp(np.sum(coeff))
31    signs = (-1.0) ** np.arange(len(coeff))
32    right_speed = np.exp(np.dot(signs, coeff))
33    if right_speed >= 1.0:
34        raise ValueError("finite-forward condition is violated")
35
36    return x, upper_share, left_speed, right_speed
37
```

```

38
39 def call_prices(z, x, upper_share, log_strikes):
40     """Normalized calls from asset-tail minus cash-tail."""
41     inverse = PchipInterpolator(x, z)
42     share = PchipInterpolator(z, upper_share)
43     roots = inverse(log_strikes)
44     exercise_p = expit(roots)
45     return share(roots) - np.exp(log_strikes) * (1.0 - exercise_p)

```

Two details are deliberately omitted from the listing. First, truncating z without the analytic tail corrections in appendix A biases the mean and far-wing prices. Second, a production implementation should evaluate the admissibility condition before exponentials overflow, then use scaled or log-domain accumulation near the boundary.

What to test before optimizing

For a random interior coefficient vector: refine the grid; check $\mathbb{E}[Y] = 1$; check that x is increasing; compare (22) with direct payoff quadrature; verify parity; recover density from discrete butterflies; and compare every analytic Jacobian column with symmetric finite differences. Only after those checks should a small calibration residual be treated as evidence.

References

- [1] D. T. Breeden and R. H. Litzenberger, Prices of state-contingent claims implicit in option prices, *Journal of Business*, 51(4):621–651, 1978.
- [2] R. W. Lee, The moment formula for implied volatility at extreme strikes, *Mathematical Finance*, 14(3):469–480, 2004.
- [3] S. Benaim and P. K. Friz, Regular variation and smile asymptotics, *Mathematical Finance*, 19(1):1–12, 2009.
- [4] G. H. Hardy, J. E. Littlewood, and G. Pólya, *Inequalities*, Cambridge University Press, second edition, 1952.
- [5] M. Shaked and J. G. Shanthikumar, *Stochastic Orders*, Springer, 2007.
- [6] P. S. Hagan and G. West, Interpolation methods for curve construction, *Applied Mathematical Finance*, 13(2):89–129, 2006.
- [7] F. N. Fritsch and R. E. Carlson, Monotone piecewise cubic interpolation, *SIAM Journal on Numerical Analysis*, 17(2):238–246, 1980.
- [8] J. Gatheral, *The Volatility Surface: A Practitioner’s Guide*, Wiley, 2006.
- [9] P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer, 2004.